

SUCCESS STORY **IN FOCUS!**



Mini-Project - Big Basket Sales and Product Analysis

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 1: Load DataSet

```
df=pd.read_csv("/content/BigBasket Products.csv")
df

{"summary":{"\n  \"name\": \"df\", \n  \"rows\": 27555, \n  \"fields\": [\n    {\n      \"column\": \"index\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 7954, \n        \"min\": 1, \n        \"max\": 27555, \n        \"num_unique_values\": 27555, \n        \"samples\": [\n          16202, \n          24963, \n          18965\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      }\n    }\n  ]\n}}
```

```

}\n    },\n    {\n        \"column\": \"product\", \n        \"properties\": {\n            \"dtype\": \"string\", \n            \"num_unique_values\": 23540, \n            \"samples\": [\n                \"Pure Impact Body Wash With Purifying Micro Particles - Shower Gel For Body, Face & Hair\", \n                \"Premium Care XXL - 30 Diaper Pants\", \n                \"Tomato - Green\" \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    }, \n    {\n        \"column\": \"category\", \n        \"properties\": {\n            \"dtype\": \"category\", \n            \"num_unique_values\": 11, \n            \"samples\": [\n                \"Snacks & Branded Foods\", \n                \"Beauty & Hygiene\", \n                \"Fruits & Vegetables\" \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    }, \n    {\n        \"column\": \"sub_category\", \n        \"properties\": {\n            \"dtype\": \"category\", \n            \"num_unique_values\": 90, \n            \"samples\": [\n                \"Crocery & Cutlery\", \n                \"Fragrances & Deos\", \n                \"Energy & Soft Drinks\" \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    }, \n    {\n        \"column\": \"brand\", \n        \"properties\": {\n            \"dtype\": \"category\", \n            \"num_unique_values\": 2313, \n            \"samples\": [\n                \"Red Label\", \n                \"Lomani\", \n                \"Happily Unmarried\" \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    }, \n    {\n        \"column\": \"sale_price\", \n        \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 1202.102113074331, \n            \"min\": 2.45, \n            \"max\": 112475.0, \n            \"num_unique_values\": 3261, \n            \"samples\": [\n                24.95, \n                40.0, \n                5040.0 \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    }, \n    {\n        \"column\": \"market_price\", \n        \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 581.7307174034938, \n            \"min\": 3.0, \n            \"max\": 12500.0, \n            \"num_unique_values\": 1348, \n            \"samples\": [\n                192.0, \n                1164.0, \n                74.38 \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    }, \n    {\n        \"column\": \"type\", \n        \"properties\": {\n            \"dtype\": \"category\", \n            \"num_unique_values\": 426, \n            \"samples\": [\n                \"Nappies & Rash Cream\", \n                \"Hair Removal\", \n                \"Crackers & Digestive\" \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    }, \n    {\n        \"column\": \"rating\", \n        \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 0.7392170543515166, \n            \"min\": 1.0, \n            \"max\": 5.0, \n            \"num_unique_values\": 40, \n            \"samples\": [\n                3.0, \n                1.5, \n                3.9 \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    }, \n    {\n        \"column\": \"description\", \n        \"properties\": {\n            \"dtype\": \"string\", \n            \"num_unique_values\": 21944, \n            \"samples\": [\n                \"Premium Day Face Cream: Lightweight formula that will give your skin

```

a daily boost of radiance and hydration. Packed with the goodness of skin brightening, Vitamin C Why Is It Special?- It contains the Most Stable & Effective form of Vitamin C, 3-O Ethyl Ascorbic Acid. It is the next generation Vitamin C that really works in improving skin health.- Vitamin C is a proven skin-brightening agent known for its antioxidant properties for decades. It boosts collagen formation and effectively targets visible signs of ageing, giving skin the much-needed glow. - The presence of Vitamin E, Pro-Vitamin B5 & Vitamin B3 helps awaken and clarify your complexion, revealing a natural healthy-looking glow. - SPF 30 & PA+++ provide the highest level of protection against UVA & UVB rays of the sun. The mineral-based sun protection agents shield the skin against the harsh solar radiation, minimizing skin dryness and dullness. - Multiple natural oils are combined for intense hydration and nourishment. Antioxidant-rich botanical extracts are added to keep skin moisturized, smooth & soft. - St.Botanica is a PETA-certified cruelty-free brand. The product does not contain any harmful chemicals & is free from parabens, silicones or mineral oils.\",\n

\"Kari Kari Japanese premium snacks are the exciting irresistible new taste of crunch. Lite and crispy, the savoury rice crackers are mixed with peanuts to make hearty, wholesome snack with delicious, bold and zesty taste with every bite.\",\n

\"Safely store your food in this pink lunch box. The lunch box is made plastic which gives the container durability and sturdiness. The rectangular containers come with an airtight lid to keep food fresh for long. The wide mouth opening of the lunch makes it easy to fill food and easy to clean. The set makes a great gift option for special days.\",\n

```
],\n    \"semantic_type\": \"\",\n    \"description\": \"\"\n  }\n  },\n  {\n    \"column\": \"product\", \n    \"properties\": {\n      \"dtype\": \"string\", \n      \"num_unique_values\": 23540, \n      \"samples\": [\n        \"Pure Impact Body Wash With Purifying Micro Particles - Shower Gel For Body, Face & Hair\", \n        \"Premium Care XXL - 30 Diaper Pants\", \n        \"Tomato - Green\" \n      ], \n
```

Step 2: Use head function to look for first 12 rows.

```
df.head(12)
```

```
{\"summary\":{\"name\": \"df\", \"rows\": 27555, \"fields\": [\n  {\n    \"column\": \"index\", \n    \"properties\": {\n      \"dtype\": \"number\", \n      \"std\": 7954, \n      \"min\": 1, \n      \"max\": 27555, \n      \"num_unique_values\": 27555, \n      \"samples\": [\n        16202, \n        24963, \n        18965 \n      ], \n      \"semantic_type\": \"\", \n      \"description\": \"\" \n    }, \n    {\n      \"column\": \"product\", \n      \"properties\": {\n        \"dtype\": \"string\", \n        \"num_unique_values\": 23540, \n        \"samples\": [\n          \"Pure Impact Body Wash With Purifying Micro Particles - Shower Gel For Body, Face & Hair\", \n          \"Premium Care XXL - 30 Diaper Pants\", \n          \"Tomato - Green\" \n        ], \n
```

```

\"semantic_type\": \"\",
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      \"samples\": [
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        \"Beauty & Hygiene\",
        \"Fruits & Vegetables\"
      ],
      \"semantic_type\": \"\",
      \"description\": \"\",
    },
    {
      \"column\": \"sub_category\",
      \"properties\": {
        \"dtype\": \"category\",
        \"num_unique_values\": 90,
        \"samples\": [
          \"Crocery & Cutlery\",
          \"Fragrances & Deos\",
          \"Energy & Soft Drinks\"
        ],
        \"semantic_type\": \"\",
        \"description\": \"\",
      },
      {
        \"column\": \"brand\",
        \"properties\": {
          \"dtype\": \"category\",
          \"num_unique_values\": 2313,
          \"samples\": [
            \"Red Label\",
            \"Lomani\",
            \"Happily Unmarried\"
          ],
          \"semantic_type\": \"\",
          \"description\": \"\",
        },
        {
          \"column\": \"sale_price\",
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            \"std\": 1202.102113074331,
            \"min\": 2.45,
            \"max\": 112475.0,
            \"num_unique_values\": 3261,
            \"samples\": [
              24.95,
              40.0,
              5040.0
            ],
            \"semantic_type\": \"\",
            \"description\": \"\",
          },
          {
            \"column\": \"market_price\",
            \"properties\": {
              \"dtype\": \"number\",
              \"std\": 581.7307174034938,
              \"min\": 3.0,
              \"max\": 12500.0,
              \"num_unique_values\": 1348,
              \"samples\": [
                192.0,
                1164.0,
                74.38
              ],
              \"semantic_type\": \"\",
              \"description\": \"\",
            },
            {
              \"column\": \"type\",
              \"properties\": {
                \"dtype\": \"category\",
                \"num_unique_values\": 426,
                \"samples\": [
                  \"Nappies & Rash Cream\",
                  \"Hair Removal\",
                  \"Crackers & Digestive\"
                ],
                \"semantic_type\": \"\",
                \"description\": \"\",
              },
              {
                \"column\": \"rating\",
                \"properties\": {
                  \"dtype\": \"number\",
                  \"std\": 0.7392170543515166,
                  \"min\": 1.0,
                  \"max\": 5.0,
                  \"num_unique_values\": 40,
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                  \"description\": \"\",
                },
                {
                  \"column\": \"description\",
                  \"properties\": {
                    \"dtype\": \"string\",
                    \"num_unique_values\": 21944,
                    \"samples\": [
                      \"Premium Day Face Cream:Lightweight formula that will give your skin a daily boost of radiance and hydration. Packed with the goodness of skin brightening, Vitamin C Why Is It Special?- It contains the Most Stable & Effective form of Vitamin C, 3-0 Ethyl Ascorbic Acid. It is the next generation Vitamin C that really works in improving skin health.- Vitamin C is a proven skin-brightening agent known for its antioxidant properties for decades. It boosts collagen formation and

```

```
effectively targets visible signs of ageing, giving skin the much-
needed glow. - The presence of Vitamin E, Pro-Vitamin B5 & Vitamin B3
helps awaken and clarify your complexion, revealing a natural healthy-
looking glow. - SPF 30 & PA+++ provide the highest level of protection
against UVA & UVB rays of the sun. The mineral-based sun protection
agents shield the skin against the harsh solar radiation, minimizing
skin dryness and dullness. - Multiple natural oils are combined for
intense hydration and nourishment. Antioxidant-rich botanical extracts
are added to keep skin moisturized, smooth & soft. - St.Botanica is a
PETA-certified cruelty-free brand. The product does not contain any
harmful chemicals & is free from parabens, silicones or mineral
oils.\",\n          \"Kari Kari Japanese premium snacks are the
exciting irresistible new taste of crunch. Lite and crispy, the
savoury rice crackers are mixed with peanuts to make hearty, wholesome
snack with delicious, bold and zesty taste with every bite.\",\n
\"Safely store your food in this pink lunch box. The lunch box is made
plastic which gives the container durability and sturdiness. The
rectangular containers come with an airtight lid to keep food fresh
for long. The wide mouth opening of the lunch makes it easy to fill
food and easy to clean. The set makes a great gift option for special
days.\",\n          ],\n          \"semantic_type\": \"\", \n
\"description\": \"\" \n          } \n          ] \n
n}\", \"type\": \"dataframe\", \"variable_name\": \"df\"}
```

Step 3: Get Description of the data in the DataFrame.

```
df.describe()

{\"summary\": \"{\\n  \"name\": \"df\",\\n  \"rows\": 8,\\n  \"fields\": [\\n
\\n    \"column\": \"index\",\\n    \"properties\": {\\n
\\n      \"dtype\": \"number\",\\n      \"std\": 9930.650518628143,\\n
\\n      \"min\": 1.0,\\n      \"max\": 27555.0,\\n
\\n      \"num_unique_values\": 6,\\n      \"samples\": [\\n        27555.0,\\n
\\n        13778.0,\\n        20666.5\\n      ],\\n
\\n      \"semantic_type\": \"\",\\n      \"description\": \"\"\\n    }\\n
\\n    },\\n    {\\n      \"column\": \"sale_price\",\\n
\\n      \"properties\": {\\n        \"dtype\": \"number\",\\n        \"std\":
39430.64768451627,\\n        \"min\": 2.45,\\n        \"max\":
112475.0,\\n        \"num_unique_values\": 8,\\n        \"samples\": [\\n
334.6483912301717,\\n        190.32,\\n        27549.0\\n      ],\\n
\\n        \"semantic_type\": \"\",\\n        \"description\": \"\"\\n      }\\n
\\n      },\\n      {\\n        \"column\": \"market_price\",\\n
\\n        \"properties\": {\\n          \"dtype\": \"number\",\\n          \"std\":
9987.087169346814,\\n          \"min\": 3.0,\\n          \"max\": 27555.0,\\n
\\n          \"num_unique_values\": 8,\\n          \"samples\": [\\n
382.05666448920346,\\n          220.0,\\n          27555.0\\n        ],\\n
```

```

\"semantic_type\": \"\",
  },
  {
    \"column\": \"rating\",
    \"dtype\": \"number\",
    \"std\": 6687.726086899768,
    \"min\": 0.7392170543515166,
    \"max\": 18919.0,
    \"num_unique_values\": 8,
    \"samples\": [
      3.9432951001638563,
      4.1,
      18919.0
    ],
    \"semantic_type\": \"\",
    \"description\": \"\"
  }
],
\"type\": \"dataframe\"
}

```

#Step 4: Find Information about the DataFrame

```

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   index                 27555 non-null  int64
 1   product               27554 non-null  object
 2   category              27555 non-null  object
 3   sub_category          27555 non-null  object
 4   brand                 27554 non-null  object
 5   sale_price            27549 non-null  float64
 6   market_price          27555 non-null  float64
 7   type                  27555 non-null  object
 8   rating                18919 non-null  float64
 9   description            27440 non-null  object
dtypes: float64(3), int64(1), object(6)
memory usage: 2.1+ MB

```

#Step 5: Find out Top & least sold products

Top sold products

```

df[['product', 'category', 'sub_category']].value_counts().head(1)

```

product	category	sub_category
Turmeric Powder/Arisina Pudi Spices	Foodgrains, Oil & Masala	Masalas &

Name: count, dtype: int64

least sold products

```

df[['product', 'category', 'sub_category']].value_counts().tail(1)

```


product	category	sub_category	
pasta shell	Snacks & Branded Foods	Noodle, Pasta, Vermicelli	1

Name: count, dtype: int64

#Step 6: Measuring discount on a certain item.

```
df["Discount_percentage"]=((df["market_price"]-df["sale_price"])/
df["market_price"])*100
df["Discount_percentage"]=df["Discount_percentage"].round(2)
df[["product","market_price","sale_price","Discount_percentage",]].head(12)
```

```
{
  "summary": {
    "\n  \"name\": \"df[[\\\"product\\\"],\\\"market_price\\\"],\\\"sale_price\\\"],\\\"Discount_percentage\\\"],]]\\\",\\n  \"rows\": 12,\\n  \"fields\": [\\n    {\\n      \"column\": \"product\\\",\\n      \"properties\": {\\n        \"dtype\": \"string\\\",\\n        \"num_unique_values\": 12,\\n        \"samples\": [\\n          \"Wheat Grass Powder - Raw\\\",\\n          \"Scrub Pad - Anti- Bacterial, Regular\\\",\\n          \"Garlic Oil - Vegetarian Capsule 500 mg\\\"\\n        ],\\n        \"semantic_type\": \"\\\",\\n        \"description\": \"\\\"\\\"\\n      }\\n    },\\n    {\\n      \"column\": \"market_price\\\",\\n      \"properties\": {\\n        \"dtype\": \"number\\\",\\n        \"std\": 291.31440632645223,\\n        \"min\": 20.0,\\n        \"max\": 1098.0,\\n        \"num_unique_values\": 11,\\n        \"samples\": [\\n          199.0,\\n          220.0,\\n          290.0\\n        ],\\n        \"semantic_type\": \"\\\",\\n        \"description\": \"\\\"\\\"\\n      }\\n    },\\n    {\\n      \"column\": \"sale_price\\\",\\n      \"properties\": {\\n        \"dtype\": \"number\\\",\\n        \"std\": 297.2815724752097,\\n        \"min\": 20.0,\\n        \"max\": 1098.0,\\n        \"num_unique_values\": 12,\\n        \"samples\": [\\n          261.0,\\n          20.0,\\n          220.0\\n        ],\\n        \"semantic_type\": \"\\\",\\n        \"description\": \"\\\"\\\"\\n      }\\n    },\\n    {\\n      \"column\": \"Discount_percentage\\\",\\n      \"properties\": {\\n        \"dtype\": \"number\\\",\\n        \"std\": 15.346066419296335,\\n        \"min\": 0.0,\\n        \"max\": 52.4,\\n        \"num_unique_values\": 5,\\n        \"samples\": [\\n          52.4,\\n          10.0,\\n          15.34\\n        ],\\n        \"semantic_type\": \"\\\",\\n        \"description\": \"\\\"\\\"\\n      }\\n    }\\n  ]\\n}\",
    \"type\": \"dataframe\"
  }
}
```

#Step 7: Find out the Missing Values from the Dataset.

```
missing_value=df.isnull().sum()
missing_value
```

index	0
product	1
category	0
sub_category	0
brand	1

```
sale_price          6
market_price        0
type                0
rating              8636
description          115
Discount_percentage 6
dtype: int64
```

```
df[df["product"].isna()]
```

```
{"repr_error": "0", "type": "dataframe"}
```

```
df[df["brand"].isna()]
```

```
{"repr_error": "0", "type": "dataframe"}
```

```
df[df["sale_price"].isna()]
```

```
{"summary": "{\n  \"name\": \"df[df[\"sale_price\"]]\",\n  \"rows\": 6,\n  \"fields\": [\n    {\n      \"column\": \"index\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 351,\n        \"min\": 1720,\n        \"max\": 2403,\n        \"num_unique_values\": 6,\n        \"samples\": [\n          1720,\n          1721,\n          2403\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"product\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 6,\n        \"samples\": [\n          \"Puja Flower Wicks - Puvvu Vathulu Batti\",\n          \"Powder - Sambar\",\n          \"Klassic Plain Cocktail Napkins (22 X 22 cm)\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"category\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 4,\n        \"samples\": [\n          \"Foodgrains, Oil & Masala\",\n          \"Snacks & Branded Foods\",\n          \"Cleaning & Household\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"sub_category\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 6,\n        \"samples\": [\n          \"Pooja Needs\",\n          \"Masalas & Spices\",\n          \"Disposables, Garbage Bag\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"brand\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 6,\n        \"samples\": [\n          \"Jaji\",\n          \"Annapoorna\",\n          \"Origami\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"sale_price\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": null,\n        \"min\": null,\n        \"max\": null,\n        \"num_unique_values\": 0,\n        \"samples\": [],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\":
```



```

{"market_price": 106.99719622494787, "std": 106.99719622494787, "min": 20.0, "max": 299.0, "num_unique_values": 6, "samples": [], "semantic_type": "", "description": "", "column": "market_price", "properties": {"dtype": "number", "num_unique_values": 6, "samples": [], "semantic_type": "", "description": "", "column": "market_price", "properties": {"dtype": "number", "std": 0.09999999999999987, "min": 3.9, "max": 4.1, "num_unique_values": 3, "samples": [], "semantic_type": "", "description": "", "column": "rating", "properties": {"dtype": "number", "std": null, "min": null, "max": null, "num_unique_values": 0, "samples": [], "semantic_type": "", "description": ""}}], "type": "dataframe"}

```

```
df[df["rating"].isna()]
```

```

{"summary": {"name": "df[df[['rating']]", "rows": 8636, "fields": [{"column": "index", "properties": {"dtype": "number", "std": 7957, "min": 56, "max": 27532, "num_unique_values": 8636, "samples": [8520, 19439], "semantic_type": "", "description": ""}}, {"column": "product", "properties": {"dtype": "string", "num_unique_values": 7646, "samples": ["Coconut Cookies", "Bright I - Anti Dark Circle Cream", "Aloevera", "Juice - Guava", "Beauty & Hygiene", "Snacks & Branded Foods", "Men's Grooming", "Drinks & Beverages", "Organic Staples"], "semantic_type": "", "description": ""}}, {"column": "sub_category", "properties": {"dtype": "category", "num_unique_values": 90, "samples": ["Fruits & Vegetables", "Beauty & Hygiene", "Snacks & Branded Foods", "Men's Grooming", "Drinks & Beverages", "Organic Staples"], "semantic_type": "", "description": ""}}], "type": "dataframe"}

```

```

\"brand\",\\n      \\\"properties\\\": {\\n          \\\"dtype\\\": \\\"category\\\",\\n
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\\\"Daucy\\\",\\n          \\\"Mama\\\",\\n          \\\"Iyers \\\"\\n          ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\
n      },\\n      {\\n          \\\"column\\\": \\\"sale_price\\\",\\n
\\\"properties\\\": {\\n          \\\"dtype\\\": \\\"number\\\",\\n          \\\"std\\\":
1824.393828437359,\\n          \\\"min\\\": 2.45,\\n          \\\"max\\\":
112475.0,\\n          \\\"num_unique_values\\\": 1766,\\n          \\\"samples\\\":
[\\n          548.8,\\n          65.0,\\n          824.98\\n          ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\
n      },\\n      {\\n          \\\"column\\\": \\\"market_price\\\",\\n
\\\"properties\\\": {\\n          \\\"dtype\\\": \\\"number\\\",\\n          \\\"std\\\":
768.8466234961703,\\n          \\\"min\\\": 5.0,\\n          \\\"max\\\": 12500.0,\\n
\\\"num_unique_values\\\": 967,\\n          \\\"samples\\\": [\\n          36.25,\\
n          134.0,\\n          658.0\\n          ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\
n      },\\n      {\\n          \\\"column\\\": \\\"type\\\",\\n          \\\"properties\\\": {\\n
\\\"dtype\\\": \\\"category\\\",\\n          \\\"num_unique_values\\\": 379,\\n
\\\"samples\\\": [\\n          \\\"Bakeware Moulds, Cutters\\\",\\n
\\\"Choco & Nut Spread\\\",\\n          \\\"Gift Boxes\\\"\\n          ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\
n      },\\n      {\\n          \\\"column\\\": \\\"rating\\\",\\n          \\\"properties\\\":
{\\n          \\\"dtype\\\": \\\"number\\\",\\n          \\\"std\\\": null,\\n
\\\"min\\\": null,\\n          \\\"max\\\": null,\\n          \\\"num_unique_values\\\":
0,\\n          \\\"samples\\\": [],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n      },\\n      {\\n          \\\"column\\\":
\\\"description\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\":
\\\"string\\\",\\n          \\\"num_unique_values\\\": 7058,\\n
\\\"samples\\\": [],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n      },\\n      {\\n          \\\"column\\\":
\\\"Discount_percentage\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\":
\\\"number\\\",\\n          \\\"std\\\": 100.04631048507072,\\n          \\\"min\\\": -
8561.68,\\n          \\\"max\\\": 83.67,\\n          \\\"num_unique_values\\\":
1172,\\n          \\\"samples\\\": [],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n      }\\n  }\\n}\\n}\\n\\\",\\\"type\\\":\\\"dataframe\\\"}

```

```
df[df[\"description\"].isna()]
```

```
{\"repr_error\":\"0\",\"type\":\"dataframe\"}
```

```
df[df[\"Discount_percentage\"].isna()]
```

```

{\\\"summary\\\":{\\n  \\\"name\\\": \\\"df[df[\\\\\\\\\\\"Discount_percentage\\\\\\\\\\\"]\\\",\\n
\\\"rows\\\": 6,\\n  \\\"fields\\\": [\\n    {\\n      \\\"column\\\": \\\"index\\\",\\n
\\\"properties\\\": {\\n      \\\"dtype\\\": \\\"number\\\",\\n      \\\"std\\\":
351,\\n      \\\"min\\\": 1720,\\n      \\\"max\\\": 2403,\\n
\\\"num_unique_values\\\": 6,\\n      \\\"samples\\\": [\\n      1720,\\n
1721,\\n      2403\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\":
\\\"product\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\": \\\"string\\\",\\n

```

```

n      \ "num_unique_values\ ": 6,\n      \ "samples\ ": [\n
\ "Puja Flower Wicks - Puvvu Vathulu Batti\ ",\n      \ "Powder -
Sambar\ ",\n      \ "Klassic Plain Cocktail Napkins (22 X 22 cm)\ "\n
],\n      \ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\ "\n
}\n      },\n      {\n      \ "column\ ": \ "category\ ",\n
\ "properties\ ": {\n      \ "dtype\ ": \ "string\ ",\n
\ "num_unique_values\ ": 4,\n      \ "samples\ ": [\n
\ "Foodgrains, Oil & Masala\ ",\n      \ "Snacks & Branded Foods\ ",\n
\ "Cleaning & Household\ "\n      ],\n      \ "semantic_type\ ":
\ "\",\n      \ "description\ ": \ "\ "\n      }\n      },\n      {\n
\ "column\ ": \ "sub_category\ ",\n      \ "properties\ ": {\n
\ "dtype\ ": \ "string\ ",\n      \ "num_unique_values\ ": 6,\n
\ "samples\ ": [\n      \ "Pooja Needs\ ",\n      \ "Masalas &
Spices\ ",\n      \ "Disposables, Garbage Bag\ "\n      ],\n
\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\ "\n      }\n
      },\n      {\n      \ "column\ ": \ "brand\ ",\n      \ "properties\ ": {\n
      \ "dtype\ ": \ "string\ ",\n      \ "num_unique_values\ ": 6,\n
\ "samples\ ": [\n      \ "Jaji\ ",\n      \ "Annapoorna\ ",\n
\ "Origami\ "\n      ],\n      \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\ "\n      }\n      },\n      {\n      \ "column\ ":
\ "sale_price\ ",\n      \ "properties\ ": {\n      \ "dtype\ ":
\ "number\ ",\n      \ "std\ ": null,\n      \ "min\ ": null,\n
\ "max\ ": null,\n      \ "num_unique_values\ ": 0,\n
\ "samples\ ": [],\n      \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\ "\n      }\n      },\n      {\n      \ "column\ ":
\ "market_price\ ",\n      \ "properties\ ": {\n      \ "dtype\ ":
\ "number\ ",\n      \ "std\ ": 106.99719622494787,\n      \ "min\ ":
20.0,\n      \ "max\ ": 299.0,\n      \ "num_unique_values\ ": 6,\n
\ "samples\ ": [],\n      \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\ "\n      }\n      },\n      {\n      \ "column\ ":
\ "type\ ",\n      \ "properties\ ": {\n      \ "dtype\ ": \ "string\ ",\n
\ "num_unique_values\ ": 6,\n      \ "samples\ ": [],\n
\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\ "\n      }\n
      },\n      {\n      \ "column\ ": \ "rating\ ",\n      \ "properties\ ":
{\n      \ "dtype\ ": \ "number\ ",\n      \ "std\ ":
0.099999999999999987,\n      \ "min\ ": 3.9,\n      \ "max\ ": 4.1,\n
\ "num_unique_values\ ": 3,\n      \ "samples\ ": [],\n
\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\ "\n      }\n
      },\n      {\n      \ "column\ ": \ "description\ ",\n
\ "properties\ ": {\n      \ "dtype\ ": \ "string\ ",\n
\ "num_unique_values\ ": 6,\n      \ "samples\ ": [],\n
\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\ "\n      }\n
      },\n      {\n      \ "column\ ": \ "Discount_percentage\ ",\n
\ "properties\ ": {\n      \ "dtype\ ": \ "number\ ",\n      \ "std\ ":
null,\n      \ "min\ ": null,\n      \ "max\ ": null,\n
\ "num_unique_values\ ": 0,\n      \ "samples\ ": [],\n
\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\ "\n      }\n
      }\n      ]\n      }","type":"dataframe"}

```

#Cleaning the Missing Values

```
df["sale_price"]=df["sale_price"].fillna(df["sale_price"].mean())
df["sale_price"]
```

```
0      220.00
1      180.00
2      119.00
3      149.00
4      162.00
```

```
...
27550    199.20
27551     67.50
27552    200.00
27553    396.00
27554    214.53
```

Name: sale_price, Length: 27555, dtype: float64

```
df["product"]=df["product"].fillna(df["product"].mode()[0])
df["product"]
```

```
0      Garlic Oil - Vegetarian Capsule 500 mg
1      Water Bottle - Orange
2      Brass Angle Deep - Plain, No.2
3      Cereal Flip Lid Container/Storage Jar - Assort...
4      Creme Soft Soap - For Hands & Body
```

```
...
27550    Wottagirl! Perfume Spray - Heaven, Classic
27551      Rosemary
27552    Peri-Peri Sweet Potato Chips
27553    Green Tea - Pure Original
27554    United Dreams Go Far Deodorant
```

Name: product, Length: 27555, dtype: object

```
df["brand"]=df["brand"].fillna(df["brand"].mode()[0])
df["brand"]
```

```
0      Sri Sri Ayurveda
1      Mastercook
2      Trm
3      Nakoda
4      Nivea
```

```
...
27550    Layerr
27551    Puramate
27552    FabBox
27553    Tetley
```

```
27554    United Colors Of Benetton
Name: brand, Length: 27555, dtype: object
```

```
df["rating"]=df["rating"].fillna(df["rating"].mean())
df["rating"]
```

```
0      4.1
1      2.3
2      3.4
3      3.7
4      4.4
```

```
...
27550   3.9
27551   4.0
27552   3.8
27553   4.2
27554   4.5
```

Name: rating, Length: 27555, dtype: float64

```
df["description"]=df["description"].fillna(df["description"].mode()
[0])
df["description"]
```

```
0      This Product contains Garlic Oil that is known...
1      Each product is microwave safe (without lid), ...
2      A perfect gift for all occasions, be it your m...
3      Multipurpose container with an attractive desi...
4      Nivea Creme Soft Soap gives your skin the best...
```

```
...
27550   Layerr brings you Wottagirl Classic fragrant b...
27551   Puramate rosemary is enough to transform a dis...
27552   We have taken the richness of Sweet Potatoes (...
27553   Tetley Green Tea with its refreshing pure, ori...
27554   The new mens fragrance from the United Dreams ...
```

Name: description, Length: 27555, dtype: object

```
df["Discount_percentage"]=df["Discount_percentage"].fillna(df["Discoun
t_percentage"].mean())
df["Discount_percentage"]
```

```
0      0.00
1      0.00
2     52.40
3     15.34
4      0.00
```

```
...
27550   20.00
27551   10.00
27552    0.00
27553   20.00
27554   44.99
```

Name: Discount_percentage, Length: 27555, dtype: float64

```
df.isnull().sum()
```

```

index          0
product        0
category       0
sub_category   0
brand          0
sale_price     0
market_price   0
type           0
rating         0
description     0
Discount_percentage  0
dtype: int64

```

```
df=pd.DataFrame(df)
```

```
df
```

```

{"summary":{"\n  \"name\": \"df\", \n  \"rows\": 27555, \n  \"fields\": [\n    {\n      \"column\": \"index\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 7954, \n        \"min\": 1, \n        \"max\": 27555, \n        \"num_unique_values\": 27555, \n        \"samples\": [\n          16202, \n          24963, \n          18965\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"product\", \n      \"properties\": {\n        \"dtype\": \"string\", \n        \"num_unique_values\": 23540, \n        \"samples\": [\n          \"Pure Impact Body Wash With Purifying Micro Particles - Shower Gel For Body, Face & Hair\", \n          \"Premium Care XXL - 30 Diaper Pants\", \n          \"Tomato - Green\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"category\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 11, \n        \"samples\": [\n          \"Snacks & Branded Foods\", \n          \"Beauty & Hygiene\", \n          \"Fruits & Vegetables\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"sub_category\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 90, \n        \"samples\": [\n          \"Crocery & Cutlery\", \n          \"Fragrances & Deos\", \n          \"Energy & Soft Drinks\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"brand\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 2313, \n        \"samples\": [\n          \"Red Label\", \n          \"Lomani\", \n          \"Happily Unmarried\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"sale_price\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 1201.9712245397184, \n        \"min\": 2.45, \n        \"max\": 112475.0, \n        \"num_unique_values\": 3262, \n        \"samples\": [\n          2310.0, \n          40.0, \n

```



```

436.5\n        ],\n        \"semantic_type\": \"\",\n\"description\": \"\"\n    }\n    },\n    {\n        \"column\":\n\"market_price\",\n        \"properties\": {\n            \"dtype\":\n\"number\",\n            \"std\": 581.7307174034938,\n            \"min\":\n3.0,\n            \"max\": 12500.0,\n            \"num_unique_values\": 1348,\n            \"samples\": [\n                192.0,\n                1164.0,\n                74.38\n            ],\n            \"semantic_type\": \"\",\n\"description\": \"\"\n    }\n    },\n    {\n        \"column\":\n\"type\",\n        \"properties\": {\n            \"dtype\": \"category\",\n            \"num_unique_values\": 426,\n            \"samples\": [\n                \"Nappies & Rash Cream\",\n                \"Hair Removal\",\n                \"Crackers & Digestive\"\n            ],\n            \"semantic_type\":\n\"\",\n            \"description\": \"\"\n    }\n    },\n    {\n        \"column\": \"rating\",\n        \"properties\": {\n            \"dtype\":\n\"number\",\n            \"std\": 0.6125157513074951,\n            \"min\":\n1.0,\n            \"max\": 5.0,\n            \"num_unique_values\": 41,\n            \"samples\": [\n                4.8,\n                4.2,\n                3.5\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n    }\n    },\n    {\n        \"column\": \"description\",\n        \"properties\": {\n            \"dtype\": \"string\",\n            \"num_unique_values\": 21944,\n            \"samples\": [\n                \"Premium Day Face Cream:Lightweight formula that will give your skin\na daily boost of radiance and hydration. Packed with the goodness of\nskin brightening, Vitamin C Why Is It Special?- It contains the Most\nStable & Effective form of Vitamin C, 3-0 Ethyl Ascorbic Acid. It is\nthe next generation Vitamin C that really works in improving skin\nhealth.- Vitamin C is a proven skin-brightening agent known for its\nantioxidant properties for decades. It boosts collagen formation and\neffectively targets visible signs of ageing, giving skin the much-\nneeded glow. - The presence of Vitamin E, Pro-Vitamin B5 & Vitamin B3\nhelps awaken and clarify your complexion, revealing a natural healthy-\nlooking glow. - SPF 30 & PA+++ provide the highest level of protection\nagainst UVA & UVB rays of the sun. The mineral-based sun protection\nagents shield the skin against the harsh solar radiation, minimizing\nskin dryness and dullness. - Multiple natural oils are combined for\nintense hydration and nourishment. Antioxidant-rich botanical extracts\nare added to keep skin moisturized, smooth & soft. - St.Botanica is a\nPETA-certified cruelty-free brand. The product does not contain any\nharmful chemicals & is free from parabens, silicones or mineral\noils.\",\n                \"Kari Kari Japanese premium snacks are the\nexciting irresistible new taste of crunch. Lite and crispy, the\nsavoury rice crackers are mixed with peanuts to make hearty, wholesome\nsnack with delicious, bold and zesty taste with every bite.\",\n                \"Safely store your food in this pink lunch box. The lunch box is made\nplastic which gives the container durability and sturdiness. The\nrectangular containers come with an airtight lid to keep food fresh\nfor long. The wide mouth opening of the lunch makes it easy to fill\nfood and easy to clean. The set makes a great gift option for special\ndays.\",\n            ],\n            \"semantic_type\": \"\",

```

```

\"description\": \"\\\"\\n      }\n    },\n    {\n      \"column\":
\"Discount_percentage\", \n      \"properties\": {\n        \"dtype\":
\"number\", \n        \"std\": 420.56480309629524, \n        \"min\": -
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2667, \n        \"samples\": [\n          20.03, \n          33.16, \n
4.05\n        ], \n        \"semantic_type\": \"\\\", \n
\"description\": \"\\\"\\n      }\n    }\n  ]\n
n}\", \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

#Step 8: Find out the outliers from the dataset according to the columns

```

numeric=df.select_dtypes(include=[\"int\", \"float\"])\nnumeric

{\"summary\": {\"\\n  \"name\": \"numeric\", \n  \"rows\": 27555, \n
\"fields\": [\n    {\n      \"column\": \"index\", \n
\"properties\": {\n        \"dtype\": \"number\", \n        \"std\":
7954, \n        \"min\": 1, \n        \"max\": 27555, \n
\"num_unique_values\": 27555, \n        \"samples\": [\n
16202, \n        24963, \n        18965\n      ], \n
\"semantic_type\": \"\\\", \n        \"description\": \"\\\"\\n      }\n
    }, \n    {\n      \"column\": \"sale_price\", \n
\"properties\": {\n        \"dtype\": \"number\", \n        \"std\":
1201.9712245397184, \n        \"min\": 2.45, \n        \"max\":
112475.0, \n        \"num_unique_values\": 3262, \n        \"samples\":
[\n          2310.0, \n          40.0, \n          436.5\n        ], \n
\"semantic_type\": \"\\\", \n        \"description\": \"\\\"\\n      }\n
    }, \n    {\n      \"column\": \"market_price\", \n
\"properties\": {\n        \"dtype\": \"number\", \n        \"std\":
581.7307174034938, \n        \"min\": 3.0, \n        \"max\": 12500.0, \n
\"num_unique_values\": 1348, \n        \"samples\": [\n
192.0, \n        1164.0, \n        74.38\n      ], \n
\"semantic_type\": \"\\\", \n        \"description\": \"\\\"\\n      }\n
    }, \n    {\n      \"column\": \"rating\", \n        \"properties\":
{\n          \"dtype\": \"number\", \n          \"std\":
0.6125157513074951, \n          \"min\": 1.0, \n          \"max\": 5.0, \n
\"num_unique_values\": 41, \n          \"samples\": [\n
4.8, \n          4.2, \n          3.5\n        ], \n          \"semantic_type\": \"\\\", \n
\"description\": \"\\\"\\n      }\n    }, \n    {\n      \"column\":
\"Discount_percentage\", \n      \"properties\": {\n        \"dtype\":
\"number\", \n        \"std\": 420.56480309629524, \n        \"min\": -
68813.95, \n        \"max\": 83.67, \n        \"num_unique_values\":
2667, \n        \"samples\": [\n          20.03, \n          33.16, \n
4.05\n        ], \n        \"semantic_type\": \"\\\", \n
\"description\": \"\\\"\\n      }\n    }\n  ]\n
n}\", \"type\": \"dataframe\", \"variable_name\": \"numeric\"}

def select_outliers(data):
    q1=data.quantile(.25)

```

```

q3=data.quantile(.75)
iqr=q3-q1
upper=q3+1.5*iqr
lower=q1-1.5*iqr
outliers=data[(data<lower)|(data>upper)]
return outliers.count()

numeric_colm=[ "sale_price"      ,"market_price" ,"rating"  ,"Discount
_percentage"]
outliers_count={col:select_outliers(df[col])for col in numeric_colm}
outliers_count

{'sale_price': 2205,
 'market_price': 2147,
 'rating': 5925,
 'Discount_percentage': 438}

```

#CLEANING DATA AND REPLACE THE OUTLIERS WITH HELP OF MEAN

```

def replace_outliers(df,colm):
    q1=df[colm].quantile(.25)
    q3=df[colm].quantile(.75)
    iqr=q3-q1
    upper=q3+1.5*iqr
    lower=q1-1.5*iqr
    df.loc[(df[colm]<lower)|(df[colm]>upper),colm]=df[colm].mean()
    return df

df1=replace_outliers(df,"sale_price")
df1

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 27555,\n  \"fields\": [\n    {\n      \"column\": \"index\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 7954,\n        \"min\": 1,\n        \"max\": 27555,\n        \"num_unique_values\": 27555,\n        \"samples\": [\n          16202,\n          24963,\n          18965\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"product\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 23540,\n        \"samples\": [\n          \"Pure Impact Body Wash With Purifying Micro Particles - Shower Gel For Body, Face & Hair\",\n          \"Premium Care XXL - 30 Diaper Pants\",\n          \"Tomato - Green\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"category\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 11,\n        \"samples\": [\n          \"Snacks & Branded Foods\",\n          \"Beauty & Hygiene\",\n          \"Fruits & Vegetables\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"sub_category\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 11,\n        \"samples\": [\n          \"Beauty & Hygiene\",\n          \"Fruits & Vegetables\",\n          \"Snacks & Branded Foods\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}}

```

```

\"properties\": {\n          \"dtype\": \"category\", \n          \"num_unique_values\": 90, \n          \"samples\": [\n            \"Crockery & Cutlery\", \n            \"Fragrances & Deos\", \n            \"Energy & Soft Drinks\" \n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        }, \n        {\n          \"column\": \"brand\", \n          \"properties\": {\n            \"dtype\": \"category\", \n            \"num_unique_values\": 2313, \n            \"samples\": [\n              \"Red Label\", \n              \"Lomani\", \n              \"Happily Unmarried\" \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n          }, \n          {\n            \"column\": \"sale_price\", \n            \"properties\": {\n              \"dtype\": \"number\", \n              \"std\": 162.31945333937762, \n              \"min\": 2.45, \n              \"max\": 755.0, \n              \"num_unique_values\": 2432, \n              \"samples\": [\n                131.75, \n                469.64, \n                166.25 \n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\" \n            }, \n            {\n              \"column\": \"market_price\", \n              \"properties\": {\n                \"dtype\": \"number\", \n                \"std\": 581.7307174034938, \n                \"min\": 3.0, \n                \"max\": 12500.0, \n                \"num_unique_values\": 1348, \n                \"samples\": [\n                  192.0, \n                  1164.0, \n                  74.38 \n                ], \n                \"semantic_type\": \"\", \n                \"description\": \"\" \n              }, \n              {\n                \"column\": \"type\", \n                \"properties\": {\n                  \"dtype\": \"category\", \n                  \"num_unique_values\": 426, \n                  \"samples\": [\n                    \"Nappies & Rash Cream\", \n                    \"Hair Removal\", \n                    \"Crackers & Digestive\" \n                  ], \n                  \"semantic_type\": \"\", \n                  \"description\": \"\" \n                }, \n                {\n                  \"column\": \"rating\", \n                  \"properties\": {\n                    \"dtype\": \"number\", \n                    \"std\": 0.6125157513074951, \n                    \"min\": 1.0, \n                    \"max\": 5.0, \n                    \"num_unique_values\": 41, \n                    \"samples\": [\n                      4.8, \n                      4.2, \n                      3.5 \n                    ], \n                    \"semantic_type\": \"\", \n                    \"description\": \"\" \n                  }, \n                  {\n                    \"column\": \"description\", \n                    \"properties\": {\n                      \"dtype\": \"string\", \n                      \"num_unique_values\": 21944, \n                      \"samples\": [\n                        \"Premium Day Face Cream:Lightweight formula that will give your skin a daily boost of radiance and hydration. Packed with the goodness of skin brightening, Vitamin C Why Is It Special?- It contains the Most Stable & Effective form of Vitamin C, 3-0 Ethyl Ascorbic Acid. It is the next generation Vitamin C that really works in improving skin health.- Vitamin C is a proven skin-brightening agent known for its antioxidant properties for decades. It boosts collagen formation and effectively targets visible signs of ageing, giving skin the much-needed glow. - The presence of Vitamin E, Pro-Vitamin B5 & Vitamin B3 helps awaken and clarify your complexion, revealing a natural healthy-looking glow. - SPF 30 & PA+++ provide the highest level of protection against UVA & UVB rays of the sun. The mineral-based sun protection agents shield the skin against the harsh solar radiation, minimizing skin dryness and dullness. - Multiple natural oils are combined for

```

intense hydration and nourishment. Antioxidant-rich botanical extracts are added to keep skin moisturized, smooth & soft. - St.Botanica is a PETA-certified cruelty-free brand. The product does not contain any harmful chemicals & is free from parabens, silicones or mineral oils.

"Kari Kari Japanese premium snacks are the exciting irresistible new taste of crunch. Lite and crispy, the savoury rice crackers are mixed with peanuts to make hearty, wholesome snack with delicious, bold and zesty taste with every bite.

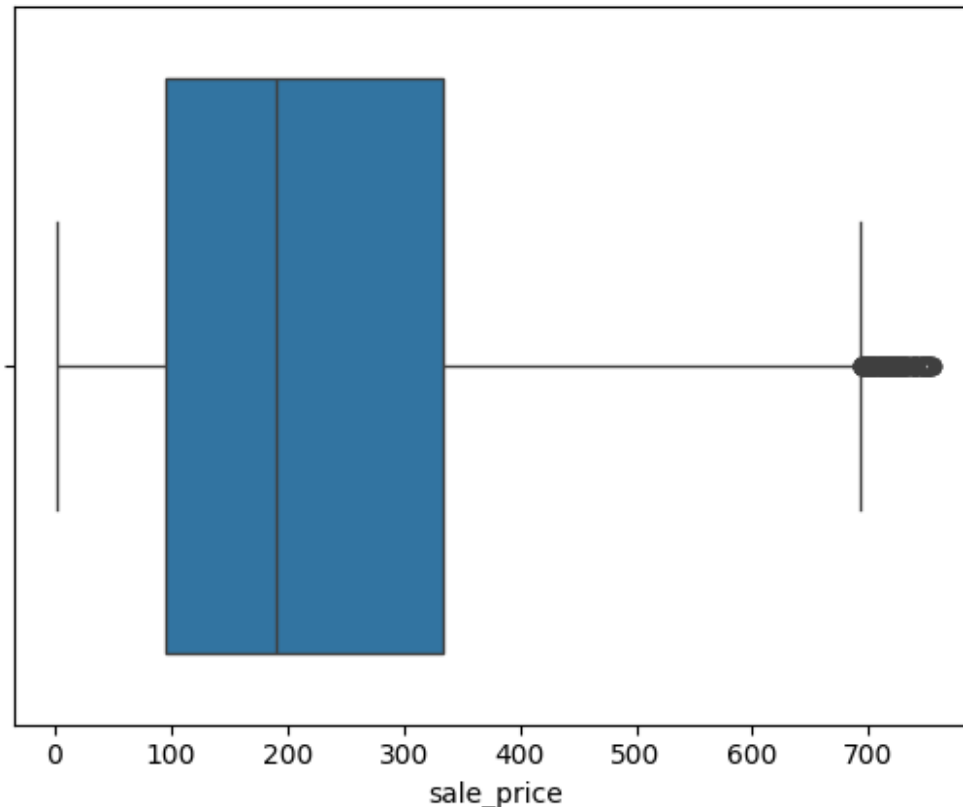
"Safely store your food in this pink lunch box. The lunch box is made plastic which gives the container durability and sturdiness. The rectangular containers come with an airtight lid to keep food fresh for long. The wide mouth opening of the lunch makes it easy to fill food and easy to clean. The set makes a great gift option for special days.

```

],
    "semantic_type": "",
    "description": "",
    },
    {
        "column":
        "Discount_percentage",
        "properties": {
            "dtype":
            "number",
            "std": 420.56480309629524,
            "min": -
68813.95,
            "max": 83.67,
            "num_unique_values":
2667,
            "samples": [
                20.03,
                33.16,
                4.05
            ],
            "semantic_type": "",
            "description": ""
        }
    }
],
    "type": "dataframe",
    "variable_name": "df"
}

sns.boxplot(data=df1,x="sale_price")
plt.show()

```



```
df2=replace_outliers(df,"market_price")
df2
```

```
{
  "summary": {
    "name": "df",
    "rows": 27555,
    "fields": [
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        "column": "index",
        "properties": {
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          "std": 7954,
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          "max": 27555,
          "num_unique_values": 27555,
          "samples": [
            16202,
            24963,
            18965
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "product",
        "properties": {
          "dtype": "string",
          "num_unique_values": 23540,
          "samples": [
            "Pure Impact Body Wash With Purifying Micro Particles - Shower Gel For Body, Face & Hair",
            "Premium Care XXL - 30 Diaper Pants",
            "Tomato - Green"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "category",
        "properties": {
          "dtype": "category",
          "num_unique_values": 11,
          "samples": [
            "Snacks & Branded Foods",
            "Beauty & Hygiene",
            "Fruits & Vegetables"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "sub_category",
        "properties": {
          "dtype": "category",
          "num_unique_values": 90,
          "samples": [

```

```

\"Crockery & Cutlery\",\\n          \"Fragrances & Deos\",\\n
\"Energy & Soft Drinks\"\\n          ],\\n          \"semantic_type\":
\"\",\\n          \"description\": \"\"\\n          }\\n          },\\n          {\\n
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\"Happily Unmarried\"\\n          ],\\n          \"semantic_type\": \"\",\\n
\"description\": \"\"\\n          }\\n          },\\n          {\\n          \"column\":
\"sale_price\",\\n          \"properties\": {\\n          \"dtype\":
\"number\",\\n          \"std\": 162.31945333937762,\\n          \"min\":
2.45,\\n          \"max\": 755.0,\\n          \"num_unique_values\": 2432,\\n
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166.25\\n          ],\\n          \"semantic_type\": \"\",\\n
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\"Hair Removal\",\\n          \"Crackers & Digestive\"\\n          ],\\n
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that will give your skin a daily boost of radiance and hydration.
Packed with the goodness of skin brightening, Vitamin C Why Is It
Special?- It contains the Most Stable & Effective form of Vitamin C,
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really works in improving skin health.- Vitamin C is a proven skin-
brightening agent known for its antioxidant properties for decades. It
boosts collagen formation and effectively targets visible signs of
ageing, giving skin the much-needed glow. - The presence of Vitamin E,
Pro-Vitamin B5 & Vitamin B3 helps awaken and clarify your complexion,
revealing a natural healthy-looking glow. - SPF 30 & PA+++ provide the
highest level of protection against UVA & UVB rays of the sun. The
mineral-based sun protection agents shield the skin against the harsh
solar radiation, minimizing skin dryness and dullness. - Multiple
natural oils are combined for intense hydration and nourishment.
Antioxidant-rich botanical extracts are added to keep skin
moisturized, smooth & soft. - St.Botanica is a PETA-certified cruelty-

```



```

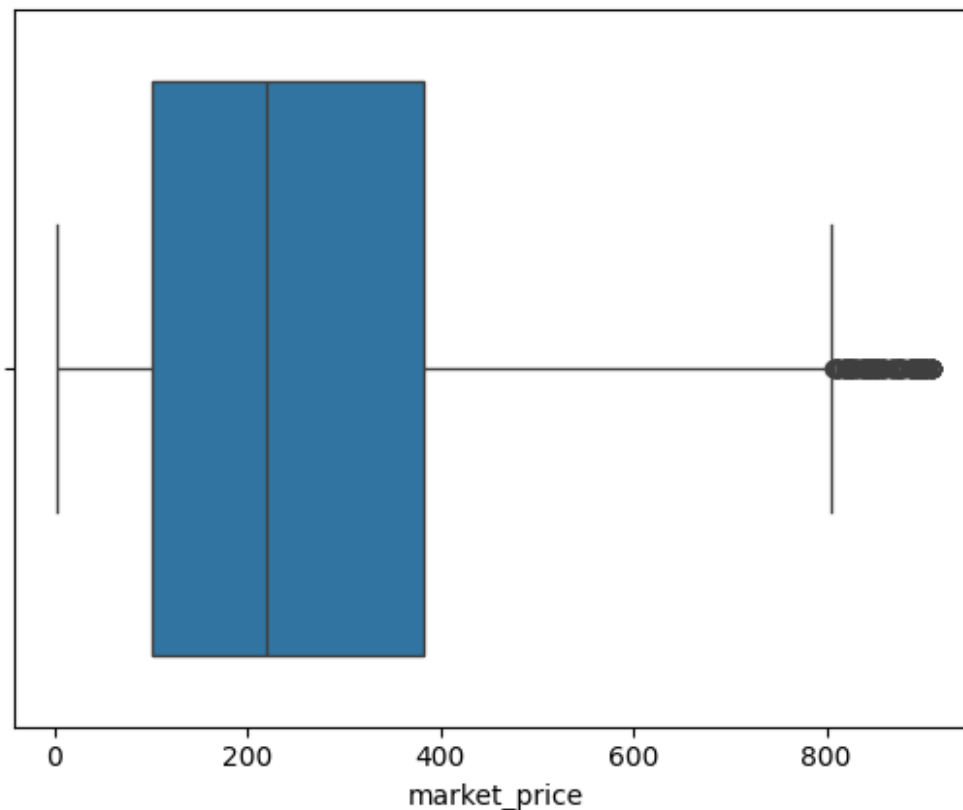
free brand. The product does not contain any harmful chemicals & is
free from parabens, silicones or mineral oils.\",\n          \nKari
Kari Japanese premium snacks are the exciting irresistible new taste
of crunch. Lite and crispy, the savoury rice crackers are mixed with
peanuts to make hearty, wholesome snack with delicious, bold and zesty
taste with every bite.\",\n          \nSafely store your food in this
pink lunch box. The lunch box is made plastic which gives the
container durability and sturdiness. The rectangular containers come
with an airtight lid to keep food fresh for long. The wide mouth
opening of the lunch makes it easy to fill food and easy to clean. The
set makes a great gift option for special days.\n\n          ],\n
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```

```

sns.boxplot(data=df2, x="market_price")
plt.show()

```



```
df3=replace_outliers(df,"rating")
df3
```

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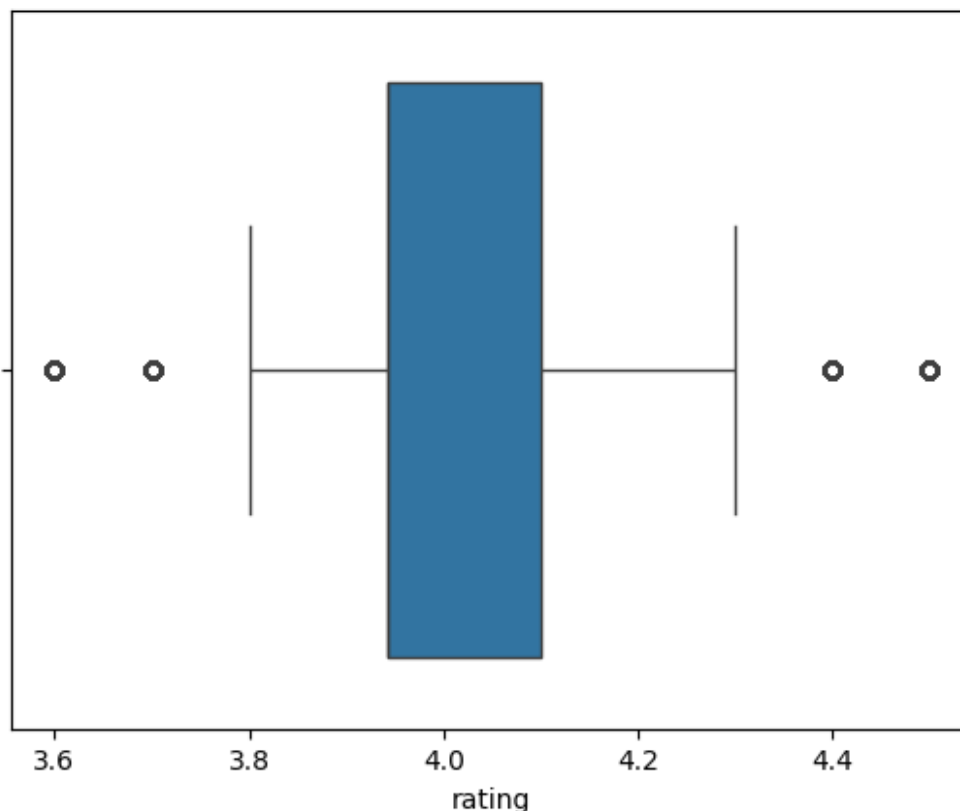
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```

```

sns.boxplot(data=df3,x="rating")
plt.show()

```



```
df4=replace_outliers(df,"Discount_percentage")
df4
```

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that will give your skin a daily boost of radiance and hydration.
Packed with the goodness of skin brightening, Vitamin C Why Is It
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```

```

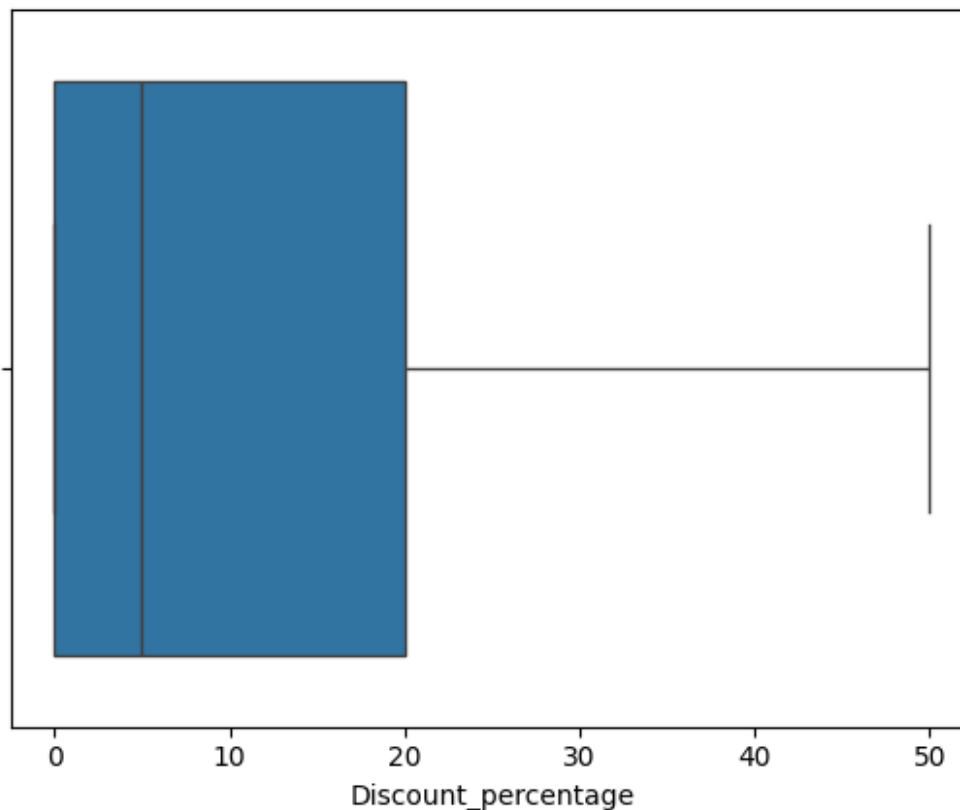
free brand. The product does not contain any harmful chemicals & is
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Kari Japanese premium snacks are the exciting irresistible new taste
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with an airtight lid to keep food fresh for long. The wide mouth
opening of the lunch makes it easy to fill food and easy to clean. The
set makes a great gift option for special days.\",\n          ],\n
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```

```

sns.boxplot(data=df4, x="Discount_percentage")
plt.show()

```



```

numeric_colm=[ "sale_price"      ,"market_price" ,"rating"  ,"Discount
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outliers_count={col:select_outliers(df[col])for col in numeric_colm}
outliers_count

{'sale_price': 365,
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df5=replace_outliers(df,"sale_price")
df5

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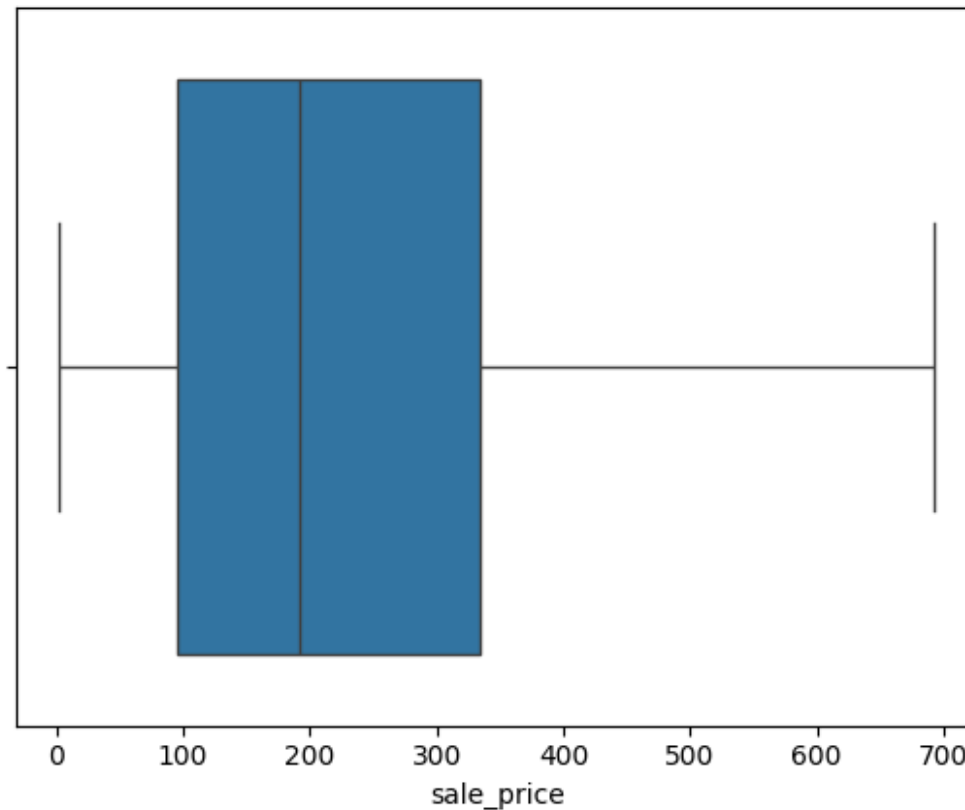
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opening of the lunch makes it easy to fill food and easy to clean. The
set makes a great gift option for special days.\", \n          ],\n
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```
sns.boxplot(data=df5,x=\"sale_price\")
plt.show()
```



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df6=replace_outliers(df,\"rating\")
df6
```

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```

helps awaken and clarify your complexion, revealing a natural healthy-looking glow. - SPF 30 & PA+++ provide the highest level of protection against UVA & UVB rays of the sun. The mineral-based sun protection agents shield the skin against the harsh solar radiation, minimizing skin dryness and dullness. - Multiple natural oils are combined for intense hydration and nourishment. Antioxidant-rich botanical extracts are added to keep skin moisturized, smooth & soft. - St.Botanica is a PETA-certified cruelty-free brand. The product does not contain any harmful chemicals & is free from parabens, silicones or mineral oils.

"Kari Kari Japanese premium snacks are the exciting irresistible new taste of crunch. Lite and crispy, the savoury rice crackers are mixed with peanuts to make hearty, wholesome snack with delicious, bold and zesty taste with every bite.

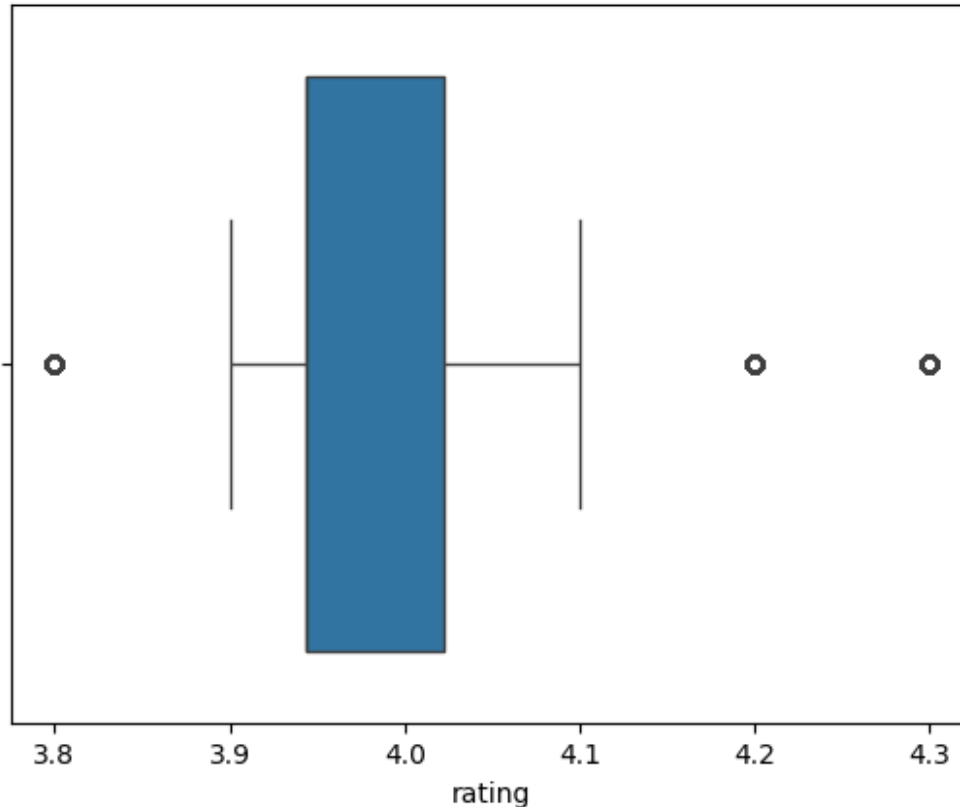
"Safely store your food in this pink lunch box. The lunch box is made plastic which gives the container durability and sturdiness. The rectangular containers come with an airtight lid to keep food fresh for long. The wide mouth opening of the lunch makes it easy to fill food and easy to clean. The set makes a great gift option for special days.

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sns.boxplot(data=df6, x="rating")
plt.show()

```



```
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```

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\"Premium Day Face Cream:Lightweight formula that will give your skin
a daily boost of radiance and hydration. Packed with the goodness of
skin brightening, Vitamin C Why Is It Special?- It contains the Most
Stable & Effective form of Vitamin C, 3-0 Ethyl Ascorbic Acid. It is
the next generation Vitamin C that really works in improving skin
health.- Vitamin C is a proven skin-brightening agent known for its
antioxidant properties for decades. It boosts collagen formation and
effectively targets visible signs of ageing, giving skin the much-
needed glow. - The presence of Vitamin E, Pro-Vitamin B5 & Vitamin B3
helps awaken and clarify your complexion, revealing a natural healthy-
looking glow. - SPF 30 & PA+++ provide the highest level of protection
against UVA & UVB rays of the sun. The mineral-based sun protection
agents shield the skin against the harsh solar radiation, minimizing
skin dryness and dullness. - Multiple natural oils are combined for
intense hydration and nourishment. Antioxidant-rich botanical extracts
are added to keep skin moisturized, smooth & soft. - St.Botanica is a

```

```

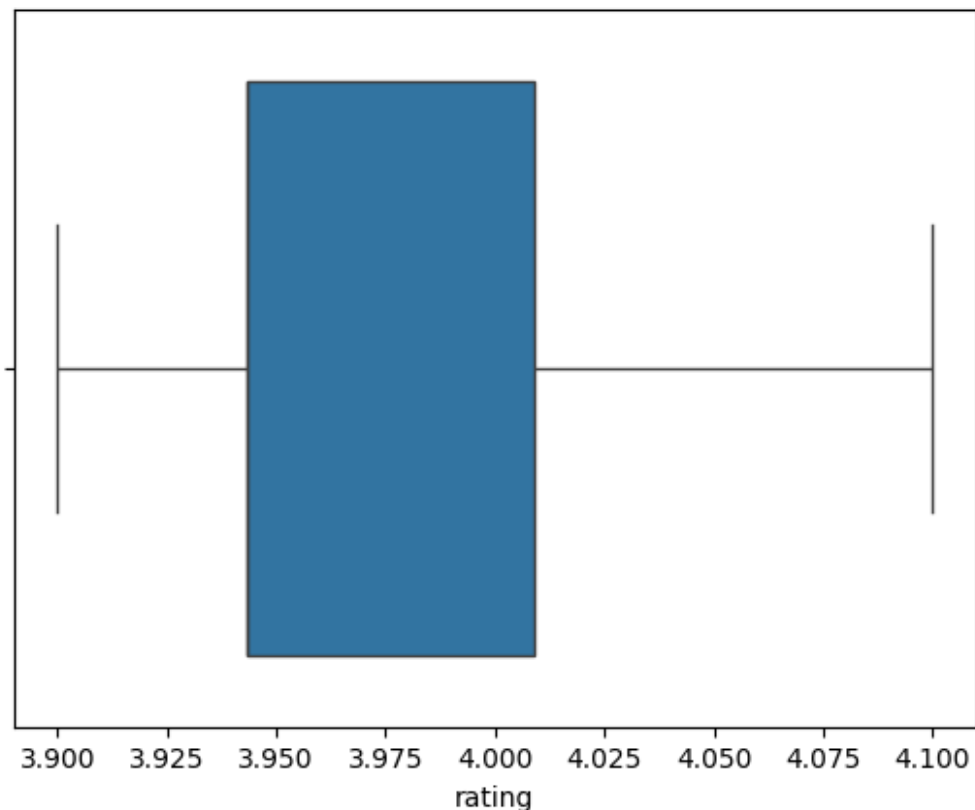
PETA-certified cruelty-free brand. The product does not contain any
harmful chemicals & is free from parabens, silicones or mineral
oils.\",\n          \"Kari Kari Japanese premium snacks are the
exciting irresistible new taste of crunch. Lite and crispy, the
savoury rice crackers are mixed with peanuts to make hearty, wholesome
snack with delicious, bold and zesty taste with every bite.\",\n
\"Safely store your food in this pink lunch box. The lunch box is made
plastic which gives the container durability and sturdiness. The
rectangular containers come with an airtight lid to keep food fresh
for long. The wide mouth opening of the lunch makes it easy to fill
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sns.boxplot(data=df7, x="rating")
plt.show()

```




```
df8=replace_outliers(df,"market_price")
df8
```

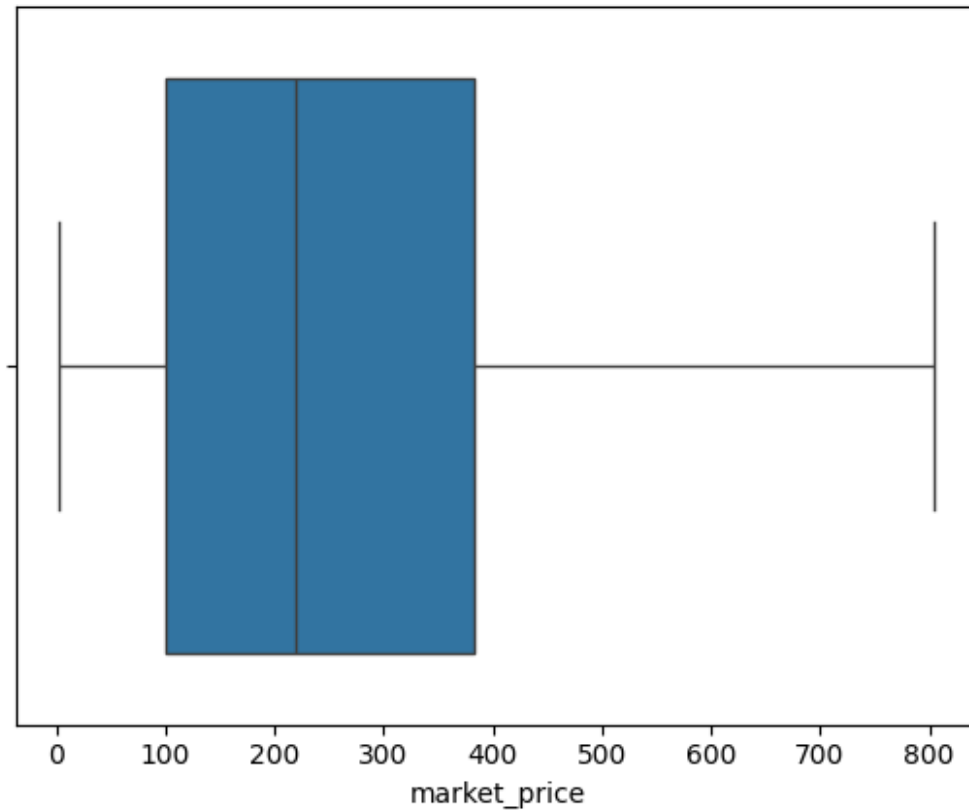
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sns.boxplot(data=df8,x="market_price")
plt.show()

```



```
numeric_colm=[ "sale_price"      ,"market_price" ,"rating"  ,"Discount
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outliers_count={col:select_outliers(df[col])for col in numeric_colm}
outliers_count

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```

#Step 9: Create Plots or visualizations.

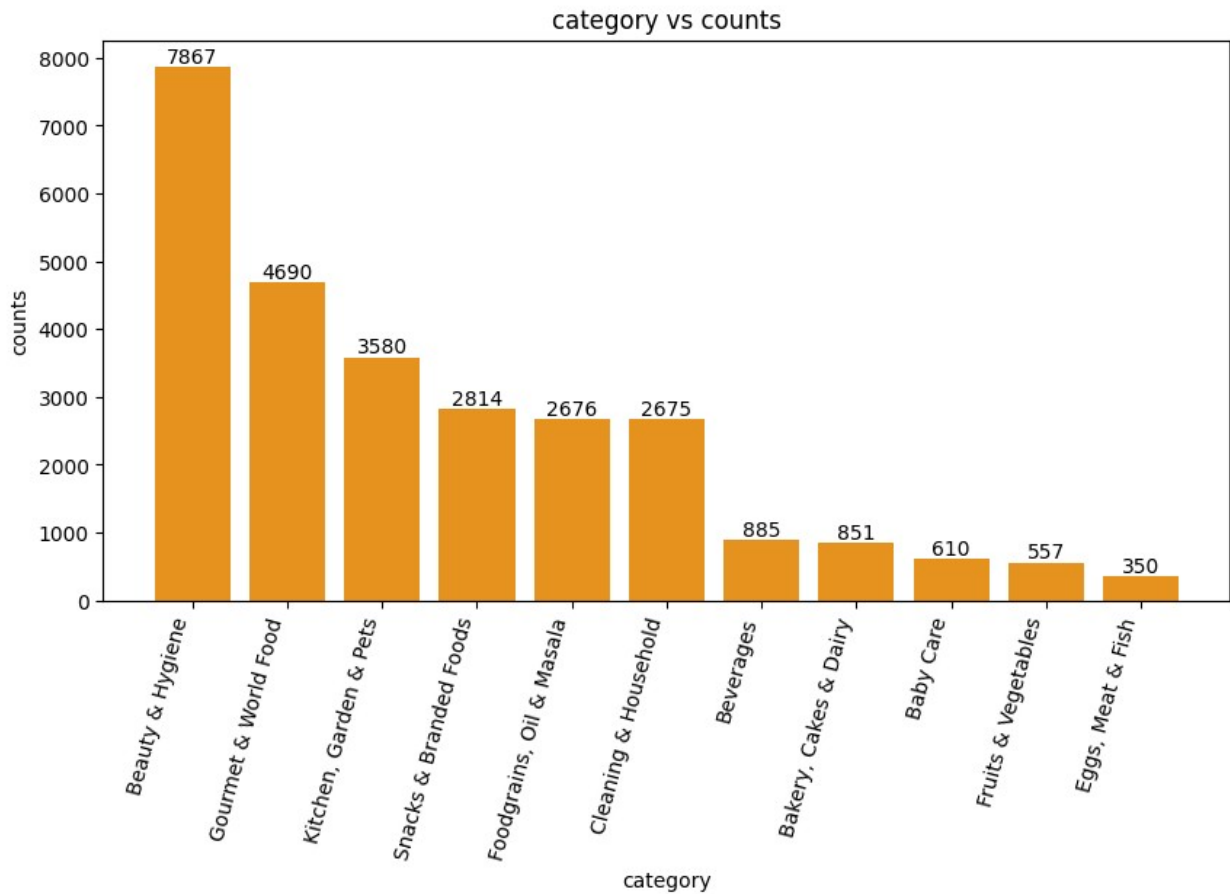
#9.1 Number of Products per Category

This bar plot shows the count of products within each category, providing insight into the distribution of products across various categories.

```
counts = df8['category'].value_counts()
counts_df = pd.DataFrame({'Category': counts.index, 'Counts':
counts.values})

plt.figure(figsize=(10,5))
plt.xticks(rotation=75, ha='right')
ax=plt.bar(counts_df["Category"],counts_df["Counts"],color="#E6921F")
plt.xlabel("category")
plt.ylabel("counts")
plt.title("category vs counts")
```

```
plt.bar_label(ax)
plt.show()
```



#Key Takeaways:

Beauty & Hygiene has the highest number of products (7,867), leading all other categories.

Gourmet & World Food ranks second with 4,690 products, followed by Kitchen, Garden & Pets with 3,580 products.

Snacks & Branded Foods, Foodgrains, Oil & Masala, and Cleaning & Household have moderate product counts ranging between 2,675 and 2,814.

Beverages, Bakery, Cakes & Dairy, and Baby Care have significantly fewer products, with counts of 885, 851, and 610, respectively.

Fruits & Vegetables and Eggs, Meat & Fish have the lowest product counts, at 557 and 350, respectively.

#Recommendations:

Capitalize on High-Performing Categories:

Invest in marketing and promotional campaigns for Beauty & Hygiene and Gourmet & World Food to further boost their performance.

Leverage their popularity to introduce new product lines or bundle offers.

Focus on Moderate-Performing Categories:

Analyze customer behavior and product performance in Snacks & Branded Foods, Foodgrains, Oil & Masala, and Cleaning & Household to identify growth opportunities.

Consider introducing premium or niche products to cater to specific customer segments.

Improve Low-Performing Categories:

Evaluate the product range and customer preferences for Bakery, Cakes & Dairy, Beverages, and Baby Care to identify gaps in offerings.

Enhance visibility and accessibility of these categories through targeted campaigns or discounts.

Expand Low-Count Categories:

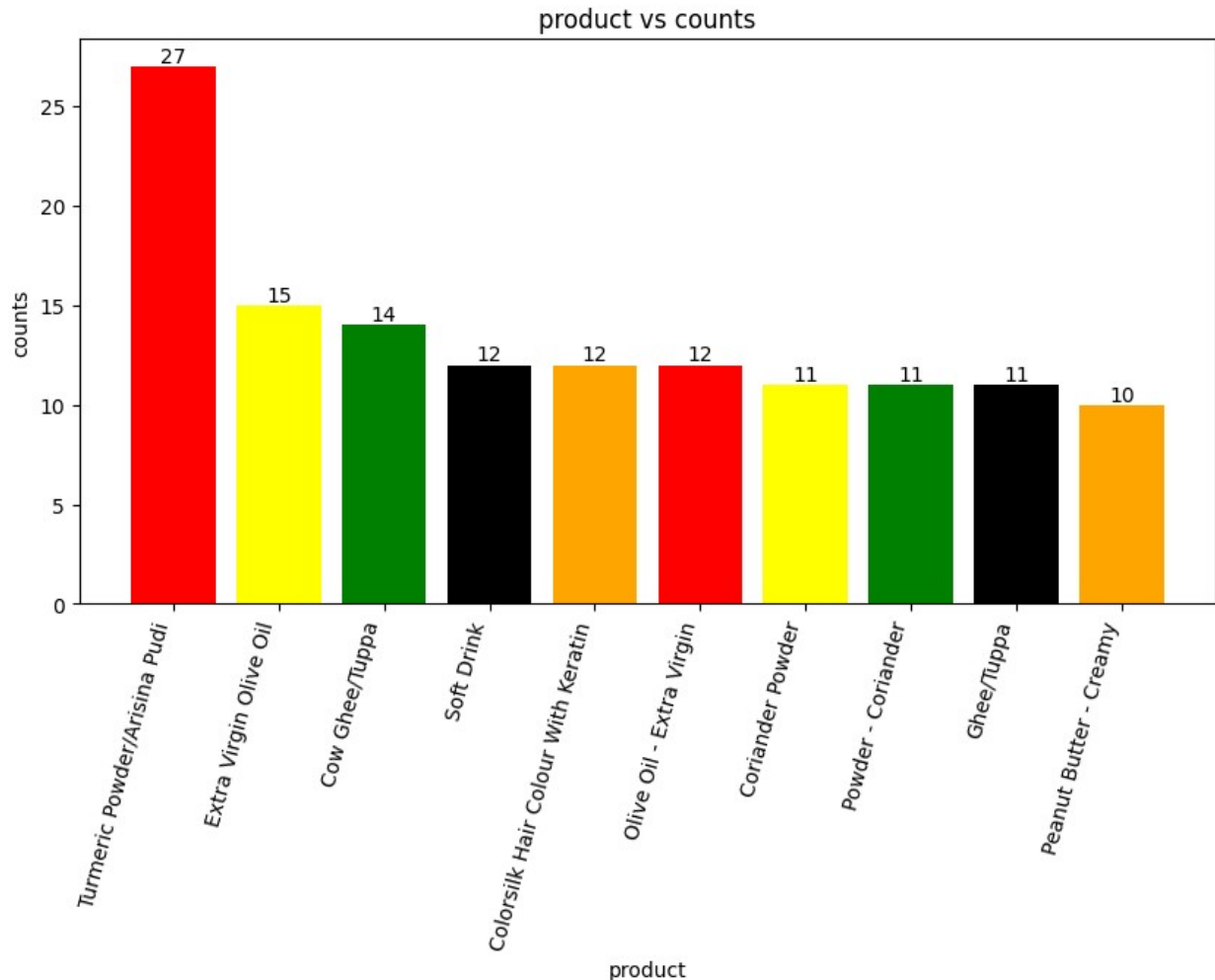
Focus on product expansion for Fruits & Vegetables and Eggs, Meat & Fish to attract a broader customer base and meet demand.

Partner with local suppliers or brands to diversify the product range in these categories.

#9.2 Top 10 Selling Products by Volume

```
counts = df8['product'].value_counts()
counts_df = pd.DataFrame({'product': counts.index, 'Counts':
counts.values})[:10]

plt.figure(figsize=(10,5))
plt.xticks(rotation=75, ha='right')
color=["red","yellow","green","black","orange"]
ax=plt.bar(counts_df["product"],counts_df["Counts"],color=color)
plt.xlabel("product")
plt.ylabel("counts")
plt.title("product vs counts")
plt.bar_label(ax)
plt.show()
```



#Key Takeaways:

Turmeric Powder/Arisina Pudi is the top-selling product with a significantly higher volume (26 units) compared to others.

Extra Virgin Olive Oil is the second-highest selling product with 15 units, followed closely by Cow Ghee/Tuppa with 14 units.

Soft Drink, Olive Oil - Extra Virgin, and Colorsilk Hair Colour With Keratin share the same volume (12 units), showcasing moderate sales performance.

Coriander Powder, Powder - Coriander, and Ghee/Tuppa each have 11 units sold, indicating steady demand.

Casting Creme Gloss Hair Color rounds out the top 10 with 10 units sold, slightly lower than the rest.

#Recommendations: Leverage Top-Selling Products:

Increase promotions and visibility for Turmeric Powder/Arisina Pudi and Extra Virgin Olive Oil to further capitalize on their popularity.

Bundle these products with related items to boost cross-selling opportunities.

Optimize Moderate-Performing Products:

Analyze factors driving the performance of products like Soft Drink and Colorsilk Hair Colour With Keratin to replicate success in similar categories.

Explore opportunities to enhance their visibility through targeted campaigns.

Enhance Sales for Lower-Performing Products:

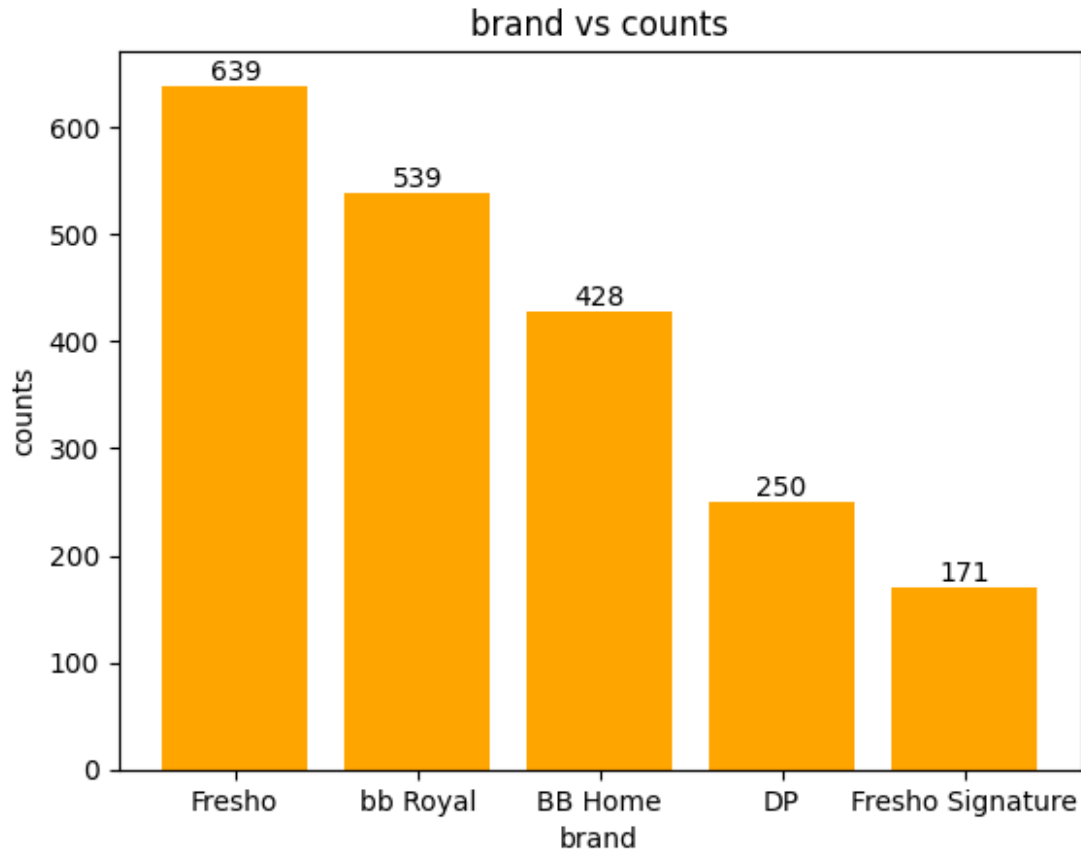
Investigate the lower sales volume of Casting Creme Gloss Hair Color and develop strategies to increase its appeal, such as offering discounts or showcasing customer testimonials.

Evaluate the product placement and pricing strategies to drive higher sales.

#9.3 Top 5 Selling Brands by Volume

```
counts = df8['brand'].value_counts()
counts_df = pd.DataFrame({'brand': counts.index, 'Counts':
counts.values})[:5]

ax=plt.bar(counts_df["brand"],counts_df["Counts"],color="orange")
plt.xlabel("brand")
plt.ylabel("counts")
plt.title("brand vs counts")
plt.bar_label(ax)
plt.show()
```



#Key Insights:

Fresho is the top-selling brand with a volume of 638 units, significantly outperforming others.

bb Royal follows with 539 units, showing strong market presence.

BB Home, DP, and Fresho Signature occupy the remaining top 5 positions, with sales volumes of 428, 250, and 171, respectively.

The volume difference between the top-performing brands (Fresho and bb Royal) and the lower performers (DP and Fresho Signature) is notable, indicating possible brand preference disparities.

#Recommendations:

Focus on Fresho and bb Royal: Strengthen marketing strategies and maintain inventory for these high-demand brands.

Improve Mid-Tier Brands: Investigate the factors limiting the sales of DP and Fresho Signature, such as pricing, placement, or promotions.

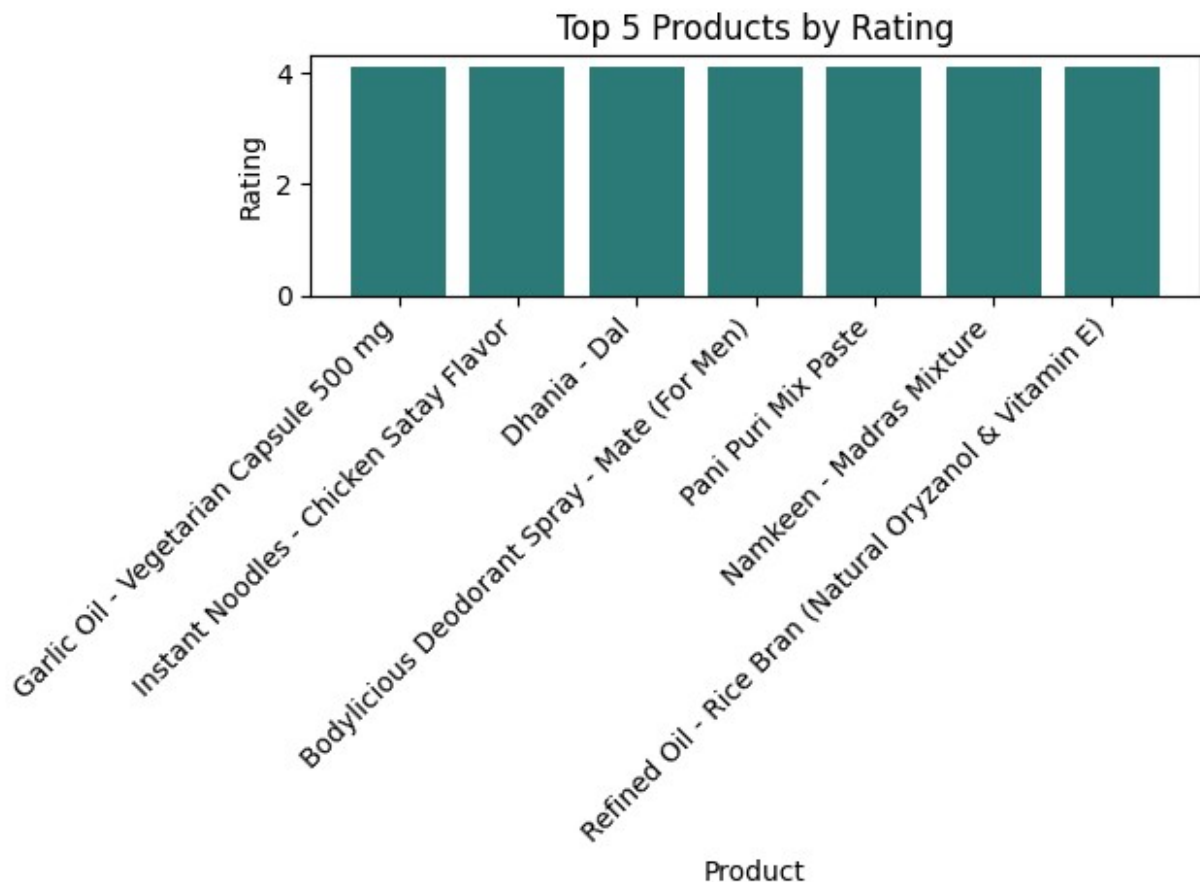
Optimize Brand Performance: Perform a detailed analysis on BB Home to identify opportunities to increase its market share further.

Customer Insights: Conduct surveys or feedback sessions to understand customer preferences for the top-selling brands. Expand Offerings: Consider introducing new products under the Fresho and bb Royal brands to capture additional market share.

#9.4 Top 5 Products by Rating:

A bar plot of the highest-rated products, showing the top 5 products with the best ratings.

```
top_5_products = df8.nlargest(7, 'rating')[['product', 'rating']]
plt.bar(top_5_products['product'], top_5_products['rating'],
color='#2B7A78')
plt.xlabel('Product')
plt.ylabel('Rating')
plt.title('Top 5 Products by Rating')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



#Key Insights:

Top 5 Products by Rating:

All five products have achieved a high rating of 4+ , indicating consistent customer satisfaction:

1. Face Wash - Oil Control, Active
2. Smooth Skin Oil - For Dry Skin
3. Salted Pumpkin
4. Veggie Cutter
5. Plain Green Olives

#Recommendations:

Maintain High Standards:

Continue to focus on quality and performance for these products to sustain high customer ratings.

Gather and showcase customer reviews for these top-rated products to build trust and attract new buyers.

Expand Related Offerings:

Introduce variations or complementary products in the same categories (e.g., more types of skincare products or kitchen tools) to capture broader interest.

Use these high-rated products as anchors to promote bundles or combos.

Leverage Marketing:

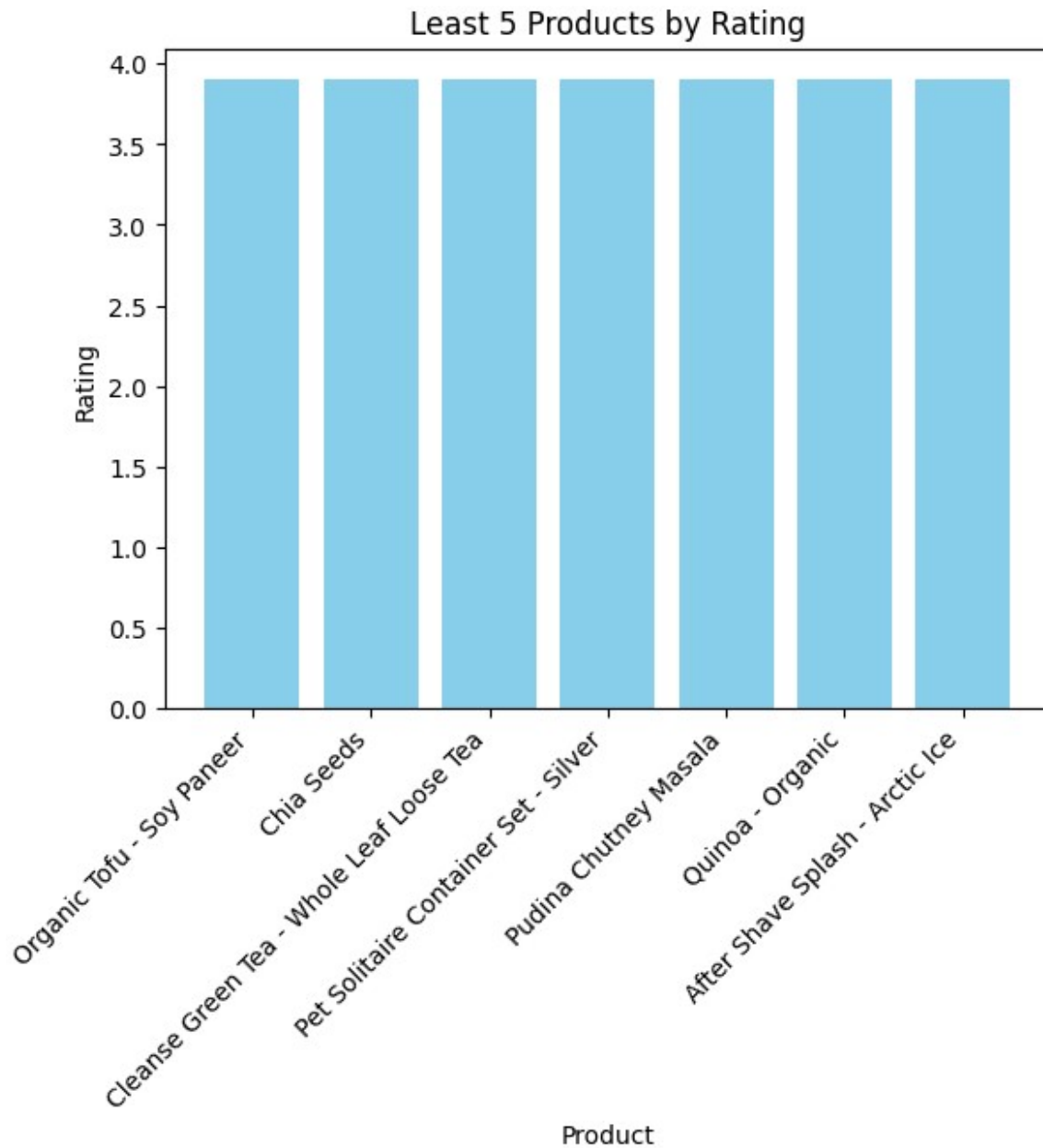
Highlight these products in promotional campaigns or feature them in "Top Rated" sections on websites or stores.

Offer limited-time discounts or loyalty rewards for these products to boost sales further.

#9.5 Least 5 Products by Rating

A bar plot of the least-rated products, showing the top 5 products with the least ratings.

```
least_5_products = df8.nsmallest(5, 'rating')[['product', 'rating']]
plt.bar(least_5_products['product'],
least_5_products['rating'],color='skyblue')
plt.xlabel('Product')
plt.ylabel('Rating')
plt.title('Least 5 Products by Rating')
plt.xticks(rotation=45, ha='right')
plt.show()
```



#Key Takeaways:

Water Bottle - Orange

Brass Angle Deep - Plain, No.2

Germ - Removal Multipurpose Wipes

Multani Mati

Biotin & Collagen Volumizing Hair Shampoo + Biotin & Collagen

Hair Conditioner

Butter Cookies Gold Collection

Insulated Hot Fresh Casserole For Roti/Chapati - White All of these products are the lowest-rated, each with a rating of 1.

#Recommendations:

Review and Improve Low-Rated Products:

Analyze the feedback for these products to identify areas for improvement, such as quality, packaging, or product description.

Consider Product Removal:

If the issues with low-rated products cannot be resolved, consider removing them from the inventory.

Gather Customer Feedback:

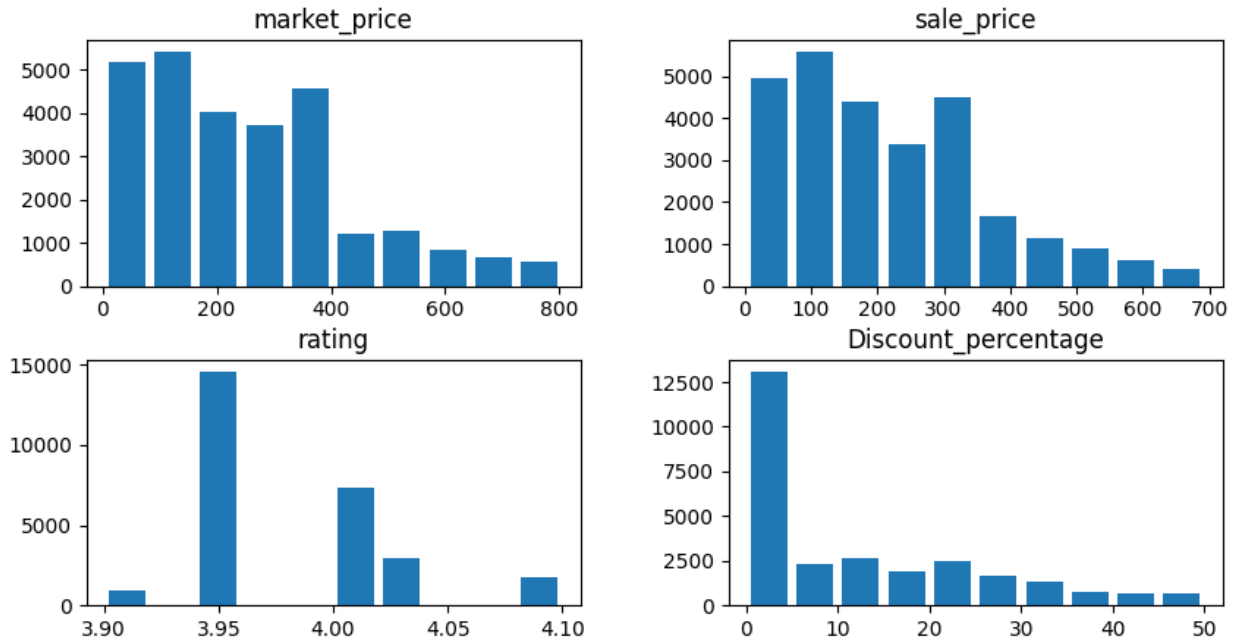
Collect feedback from customers to identify areas for improvement and maintain high ratings.

Overall Conclusion Big Basket's strategy effectively balances high-frequency essentials, strategic discounting, and in-house brand development to build customer loyalty.

Improvements could include:

Addressing quality issues for lower-rated products. Expanding the product mix in underperforming

```
df8[
["market_price", "sale_price", "rating", "Discount_percentage"]].hist(figsize=(10,5), bins=(10), rwidth=(0.8), grid=False)
plt.show()
```



```
df9=df8.select_dtypes(include=["int","float"])
df9
```

```
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```

```
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```

```
df9.corr()
```

```
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}
```

```
sns.heatmap(df9.corr(),annot=True, cmap='viridis', fmt='.2f')
plt.title('Correlation Matrix')
plt.show()
```

